# Design Brief – Review 2

## 0. Header

| | |
|---|---|
| **Project code** | FA26SE260 |
| **Project name** | CampusStay FPT – Dormitory Registration and Room-Scheduling Management System for FPT University's Tân Uyên Campus, Binh Duong |
| **Team** | CampusStay FPT — Lê Thành Minh Trung (Leader, SE183752), Nguyễn Xuân Lộc (SE183782), Trần Võ Quang Duy (SE182146), Võ Minh Luân (SE182794) |
| **Supervisor** | Thân Thị Ngọc Vân |
| **Repository link (tag review-2)** | https://github.com/tisou47/CampusStay-FPT/tree/review-2 |
| **Version** | 1.0 |
| **Date** | 28/10/2026 |

## 1. Scope

### 1.1 Registration Changes

| Section | Current | Proposed | Reason |
|---|---|---|---|
| Team, context, actors | No members; Bình Dương; Student, Staff, Admin | 4 members; Tân Uyên ward, HCMC; basis of requirements stated; + Manager (all Blocks), Staff per Block | Review 1 feedback; 2025 merger |
| Room choice, contract, check-in/out | Select room/bed; digital contract; self check-in/out | Choose Room Type and Class, engine assigns; consent to rules; desk check-in (QR pass) and check-out (damage per item); door codes | Fairness; no e-signature; keys need Staff |
| Payment, communication | VietQR payment; in-app, email, SMS | VietQR transfer + claim confirmed by a Manager; in-app, push, email | No bank API; SMS cost |
| Added scope | — | Term transition (Gia hạn, requests window, next-Term fees), registration-period requests, late check-in, Round 2, Utility and debt | Iteration 1: a dorm runs across Terms |
| Integration, technology | SIS link; forecast; Flutter | Student list import + reconciliation; dashboards; Android (Kotlin) | SIS not accessible |

### 1.2 Out of Scope

| Item | Reason |
|---|---|
| Room or Bed self-selection by students | The engine keeps gender and Block rules and fairness (D2) |
| Electronic contract signing | Needs a legal e-signature provider; consent is recorded instead |
| SMS notifications | Cost; in-app, push and email cover the cases |
| Live SIS integration, bank API, forecasting, iOS app | No access or contract; list import, reconciliation, Manager confirmation, reports and Android instead |

## 2. System at a Glance

FPT University's Tân Uyên campus will open dormitories for thousands of students and has no system to run them. CampusStay FPT takes a student from sign-in to a Bed and through every Term of the stay: Applications by Room Type and Room Class, a fair batch assignment, VietQR payments confirmed by Managers, door codes, the check-in and check-out desk, Term transitions for continuing residents, maintenance with time limits, monthly Utility billing and reports. Users: students (web and Android), Staff of each Block, dormitory Managers and the Administrator. Core value: every Bed, payment and deadline follows one set of configurable rules, with an audit

trail, instead of spreadsheets and messages. Architecture: one Spring Boot API with PostgreSQL, a React web app and an Android app (Report 4 §1.1, Figure 1).

## 3. Key Design Decisions

| ID | Decision | Options considered | Chosen option and rationale | Owner | Related FR / diagram |
|---|---|---|---|---|---|
| D1 | System structure | Microservices; modular monolith | One Spring Boot API + one PostgreSQL, layered packages per feature: Allocation and payment in one ACID transaction; one API for three clients | Lê Thành Minh Trung | SDD §1.1–1.2 |
| D2 | How Beds are given | Pick a Bed first come; Staff by hand; batch engine | Students choose Room Type and Class; a deterministic engine places by priority and fit; a Manager approves per Block: fair, no race | Lê Thành Minh Trung | FR-ASG-01..03 |
| D3 | Renewal of residents | Re-apply each Term; auto-renew | Next-Term Allocation on the same Bed at M2; requests to M3, decisions by M4, fees M6, unpaid at M7 releases the Bed | Nguyễn Xuân Lộc | FR-ASG-07, 08; act-wf03 |
| D4 | Room Class and prices | Class per Floor; prices editable | Class per Room per Term; prices copied into the Term at publication and locked once Invoices exist | Trần Võ Quang Duy | FR-INV-02, FR-CFG-05 |
| D5 | Notification channels | SMS gateway; email only | In-app always, FCM push, email via an outbox in the same transaction: no SMS cost, nothing lost, two languages | Võ Minh Luân | FR-NTF-01, 02 |

## 4. Top Risks

| ID | Risk | Impact | How we verify or mitigate | Owner |
|---|---|---|---|---|
| R1 | Dormitory not operating yet: real operation may differ | Rework after opening | Requirements from FPT's running dorms, Circular 27/2011, VNU-HCM; every price, window and rule configurable; weekly mentor review | Lê Thành Minh Trung |
| R2 | Manual claim confirmation at the peak (≈ 3,000 Deposits in 48 h) | Students wait | Claim stops the clock; bulk confirm with count and total; hourly summary | Lê Thành Minh Trung |
| R3 | One small VM and a Burstable database | Slow or down at peak | NFR-01 load test; health check and rollback by image tag; bigger VM in the runbook | Võ Minh Luân |

## 5. Known Gaps

| ID | Gap | Location | Plan | Owner | Due |
|---|---|---|---|---|---|
| G1 | Registration v1 items not built as written (self-selection, e-contract, self check-in/out, SMS, forecast, live SIS) | Registration form | Form v2 changes them (BM03) | Lê Thành Minh Trung | 31/10/2026 |
| G2 | Workflow tests TC-WF-01..05 are manual (SR-21) | Report 5 §4.2 | Run on staging; automate in CI | Võ Minh Luân | 30/11/2026 |
| G3 | NFR-01, 04, 06 not measured yet (SR-06) | Report 5 §4.4 | Load script, uptime check, run times | Võ Minh Luân | 15/11/2026 |
| G4 | No concurrency test for one Allocation per Bed per Term (SR-06) | TC-NFR-03 | Parallel-commit test | Nguyễn Xuân Lộc | 15/11/2026 |
| G5 | Android app has no automated or manual test cases (SR-08) | android/, Report 5 | UI tests of key screens | Võ Minh Luân | 30/11/2026 |
| G6 | Statuses in the schema never stored (SR-12); Term never Closed (SR-13) | V4, V6, V7; SDD §3.1 | Drop or use them; close Terms after the stay | Lê Thành Minh Trung | 15/11/2026 |

| ID | Gap | Location | Plan | Owner | Due |
|---|---|---|---|---|---|
| G7 | Round 2 run by a Manager not built; freed Beds not offered (SR-01, High) | FR-APP-07 | Manager run, pool of freed Beds, M8 | Lê Thành Minh Trung | 15/11/2026 |

## 6. Proposed Mandatory Features

| ID | Feature | Actor | Acceptance criteria (testable) | Test | Status |
|---|---|---|---|---|---|
| F-01 | Sign-in (password, Google) | All | Valid credentials give a session; the 5th wrong password locks the account; a new @fpt.edu.vn Google account gets no password until it sets one; no token → 401, wrong role → 403 | TC-F01-01..04 | Done |
| F-02 | Student accounts: verification and list import | Student, Manager | A pending account cannot apply; verify/reject notifies the student; an import previews each line and saves only valid lines | TC-F02-01..05 | Done |
| F-03 | Registration-period request and Term publication | Manager, Admin | A 2nd open request for a Term is refused; milestones out of order are named per field; publishing fixes the prices | TC-F03-01..05 | Done |
| F-04 | Apply for a Term | Student | Submit creates an Awaiting Deposit Application and a Deposit = Deposit % × Term price; a 2nd Application or Gia hạn above the maximum is refused | TC-F04-01..05 | Done |
| F-05 | VietQR payment claim and confirmation | Student, Manager | A claim stops the deadline; a rejection needs a reason and returns the time left (≥ 12 h); a Paid or Void Invoice cannot be claimed | TC-F05-01..03 | Done |
| F-06 | Review and assignment | Manager | A rejection needs a note; the same input gives the same proposal; commit creates Allocations and Balances; Staff cannot run | TC-F06-01..05 | Done |
| F-07 | Balance, forfeit and door code release | Student, Staff, System | A Balance unpaid at its deadline is forfeited; the code is released only when paid, after check-in start and with a confirmed code | TC-F07-01..05 | Done |
| F-08 | Check-in and check-out desk | Staff | Check-in is refused until the Balance is paid; check-out deducts Utility share and damage per item from the Utility Deposit | TC-F08-01..04 | Done |
| F-09 | Late check-in | Student, Staff, Manager | A request after check-in end + max days is refused; approval moves the Balance deadline; rejection needs a reason | TC-F09-01..03 | Done |
| F-10 | Next-Term requests | Student, Staff | An approved change moves the next-Term Bed; a Room Class change needs all continuing roommates; extension without debt is automatic | TC-F10-01..06 | Done |
| F-11 | Requests window, reconciliation and next-Term fees | Manager, System | M2 creates next-Term Allocations once; M3 releases un-extended Beds; fees wait for decisions and reconciliation; M7 releases unpaid Beds | TC-F11-01..06 | Done |
| F-12 | Turnover and operational moves | Staff, Manager | A move reads both meters, charges nothing and asks a new code for the old Room; after fees Staff cannot move next-Term Beds | TC-F12-01..03 | Done |
| F-13 | Maintenance with routing, time limits and escalation | Student, Staff, Manager | A request gets the routed assignee and the Block deadline (urgent < normal); other Blocks' Staff do not see it; unassigned ones escalate | TC-F13-01..05 | Done |
| F-14 | Meter readings and Utility Invoices | Staff | A reading below the previous is refused; a CSV with a bad line saves nothing; closing issues Invoices split by days, rounded up | TC-F14-01..04 | Done |
| F-15 | Utility debt and termination | System, Manager | One overdue Utility Invoice gives warning 1 once; two overdue months after the final warning terminate the stay; past move-out it is Overdue | TC-F15-01..04 | Done |

| ID | Feature | Actor | Acceptance criteria (testable) | Test | Status |
|---|---|---|---|---|---|
| F-16 | Inventory, Room Class by Term, furniture | Manager, Staff | A Room whose Beds are given keeps its class; only Managers set classes; moving an item updates conformance | TC-F16-01..04 | Done |
| F-17 | Notifications, announcements and push | All | Staff announce only to their Blocks; pinned announcements top the board; push after commit, dead tokens disabled | TC-F17-01..04 | Done |
| F-18 | Dashboards, reports, exports and search | Staff, Manager, Admin | Exports are CSV with the screen's columns and audited; search stays in the user's Blocks; Term dashboard shows milestones | TC-F18-01..04 | Done |

## 7. Non-Functional Requirements

| ID | Quality attribute | Metric | Threshold | Measurement method |
|---|---|---|---|---|
| NFR-01 | Performance | p95 of 10 main read endpoints, 200 users, base data | ≤ 1 s; errors < 1 % | Load script on a local server with production settings |
| NFR-02 | Security | Permission and scope tests passing | 100 % | Integration tests in CI |
| NFR-03 | Integrity | Duplicate Allocations of a Bed in a Term | 0 | Unique indexes + concurrency test |
| NFR-04 | Availability | Health checks answering UP | ≥ 99 % from 01/11/2026 | Uptime monitor, every 5 min |
| NFR-05 | Usability | Texts with a Vietnamese translation | 100 % | MessagesTest + scan of web routes |
| NFR-06 | Deployability | Push to healthy production; rollback | ≤ 15 min; ≤ 5 min | Actions run times; timed rollback |

## 8. Ownership Map

| Member | Role | Owned parts (2–3) | Main workflow owned | Key decisions / PRs |
|---|---|---|---|---|
| Lê Thành Minh Trung (SE183752) | Leader, PM, BA, full-stack | Applications and payments (APP, BIL claims); assignment and residence (ASG); architecture and DevOps | WF-01, WF-05 | D1, D2; ADR-17, ADR-22; v2 phases 1–8 |
| Nguyễn Xuân Lộc (SE183782) | Backend developer | Term transition (requests window, next-Term requests, fees); enrollment reconciliation | WF-03 | D3; v2 phase 6 |
| Trần Võ Quang Duy (SE182146) | Frontend developer (web) | Registration periods and the Term wizard (CFG + web); inventory, furniture and door-code screens; web UI and i18n | WF-02 | D4; v2 phases 3–4 |
| Võ Minh Luân (SE182794) | Mobile developer, tester | Android app and push (NTF); maintenance, routing and escalations (MNT, REQ); test documentation | WF-04 | D5; Android app |

# Appendix A – Defense 1.1 Targets

## A1. Main Workflows

| ID | Workflow | Actors | Trigger | Business outcome | Features | Entity states | Owner |
|---|---|---|---|---|---|---|---|
| WF-01 | Registration to residence | Student, Manager, Staff, System | A student signs in during an open registration period | The student is a resident with a door code | F-01, F-04, F-05, F-06, F-07 | Application: Awaiting Deposit → Submitted → Approved → Allocated; Allocation: Assigned – unpaid → Paid → Resident; Invoice: Unpaid → Awaiting confirmation → Paid | Lê Thành Minh Trung |
| WF-02 | Opening a registration period | Manager, Administrator, Student, System | A Manager requests a registration period | A published Term open to students | F-03, F-16, F-17 | Registration-period request: Submitted → In progress → Done; Term: Draft → Published | Trần Võ Quang Duy |
| WF-03 | Term transition of continuing residents | Student, Staff, Manager, System | M2 of the next Term | Continuing residents hold a paid Bed of the next Term; leavers check out | F-10, F-11, F-12, F-08 | Next-Term Allocation: Continuing → Resident or Cancelled; Room-Change Request: Submitted → Approved | Nguyễn Xuân Lộc |
| WF-04 | Maintenance with routing and time limits | Student, Staff, Manager, System | A resident reports a problem | The problem is fixed and confirmed within the time limit, or escalated | F-13, F-17, F-16 | Maintenance Request: Submitted → Assigned → In Progress → Resolved → Closed | Võ Minh Luân |
| WF-05 | Monthly Utility, debt and termination | Staff, Student, Manager, System | The month ends | Utility Invoices issued and collected; debtors warned or terminated | F-14, F-15, F-05, F-08 | Invoice: Unpaid → Overdue → Paid; Termination: Notice sent → Checked out / Overdue | Lê Thành Minh Trung |

## A2. Demo Scenarios

| ID | Workflow | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|---|
| DS-01 | WF-01 | Main | Stage A (registration open); se220001 eligible, no Application | 1. se220001 applies (4-bed, Standard, Gia hạn 2) and claims the Deposit transfer. 2. manager.a confirms the claim and approves the Application. 3. Stage B: se220004 (assigned, unpaid) claims the Balance; manager.a confirms. 4. staff.a1 confirms the Room code; the release job runs. | Application Submitted then Approved; se220004 is a resident (orange Bed) and sees the code; each step notified | Lê Thành Minh Trung |
| DS-02 | WF-01 | Exception | Stage A; se220003 has one rejected Deposit claim | 1. manager.a rejects se220003's second claim with a reason. | The Deposit Invoice is Void and the Application Expired; the student is told the reason; the Bed count is unchanged | Lê Thành Minh Trung |
| DS-03 | WF-02 | Main | Stage 0; manager.a, admin | 1. manager.a requests a period for the next Term and imports the excluded Rooms from the sample | Request Done and linked to the Term; manager.a sees the changed value; | Trần Võ Quang Duy |

| ID | Workflow | Type | Preconditions and data | Main steps | Expected result | Presenter |
|---|---|---|---|---|---|---|
| | | | | file. 2. admin builds the Term from the request, changes one price with a reason, completes the 7 steps and publishes. | students see the registration card and the announcement | |
| DS-04 | WF-02 | Exception | Stage 0; a request for the same Term is open | 1. manager.b files a second request for the same Term. 2. admin enters M3 before M2 in the Timeline step. | The second request is refused; the Timeline step names the wrong milestone; nothing is published | Trần Võ Quang Duy |
| DS-05 | WF-03 | Main | Stage A (requests window open); se210006 in A-505 | 1. se210006 asks for a room change. 2. staff.a1 approves it with A-603 (stage B). 3. manager.a uploads the enrolled list and resolves groups 1–2. 4. manager.a issues the next-Term fees; se210001 pays and claims. | The next-Term Allocation moves to A-603; fees issued once with count and total; se210001 is Continuing and paid | Nguyễn Xuân Lộc |
| DS-06 | WF-03 | Exception | Stage C; se200005 did not pay the next-Term fee by M7 | 1. The M7 job runs. 2. staff.a1 opens the turnover list. | The fee is Void, the next-Term Bed released, se200005 is on the leavers list with a check-out date | Nguyễn Xuân Lộc |
| DS-07 | WF-04 | Main | Stage A; se200009 in A-801, routing rules of Block A | 1. se200009 reports an urgent air-conditioner fault with a photo and the item. 2. staff.a2 (routed) starts and resolves it. 3. se200009 confirms. | Assigned at once with the urgent deadline; Closed; the item keeps a repair entry; notifications at each step | Võ Minh Luân |
| DS-08 | WF-04 | Exception | Stage A; se200010 in A-802 has a request unassigned past its limit | 1. The escalation job runs. 2. manager.a reassigns it. | One escalation in Overdue requests; after reassignment it closes once the request is back on track | Võ Minh Luân |
| DS-09 | WF-05 | Main | Stage A; staff.a1; meter sample file of Block A | 1. staff.a1 imports the readings and closes the month. 2. se200001 has an overdue Utility Invoice; the overdue job runs. 3. se200001 claims the transfer; manager.a confirms. | One Utility Invoice per resident split by days; warning 1 sent once; after payment the warning closes | Lê Thành Minh Trung |
| DS-10 | WF-05 | Exception | Stage A; se200004 terminated for Utility debt (A-703) | 1. The move-out date passes without check-out; the job runs. | Termination Overdue; Managers and Block Staff told; A-703 needs a new door code | Lê Thành Minh Trung |

## A3. Functional Test Case Target

| Source | Count | Minimum per item | Subtotal |
|---|---|---|---|
| Mandatory features | 18 | 3 | 79 |
| Main workflows (end-to-end) | 5 | 1 | 5 |
| State transitions of main entities | 68 | 1 | 68 |
| Agreed NFRs | 6 | 1 | 6 |
| Committed total | | | 158 |

Automated in CI: 138 of 158 committed test cases (87 %); the rest are the end-to-end workflow cases, the manual state transitions and the measured NFRs (Report 5 §4).